

Neural Information Retrieval

Retrieval Augmented Generation

Last lecture

- ❖ You have learned the fundamentals about Neural Retrieval models.
 - Conceptual framework for IR/NIR
 - Two architecture
 - Interaction-based
 - Representation-based
 - How to train
 - How to prepare/gather the data
 - How to optimize
 - Examples of models
 - Shallow
 - Deep

This lecture

❖ Introduction

- Language Model
- Sampling

❖ RAG

- The gap
- Architecture
- VectorDB

❖ Literature examples

- FLARE
- ITER-RETGEN
- Self-RAG

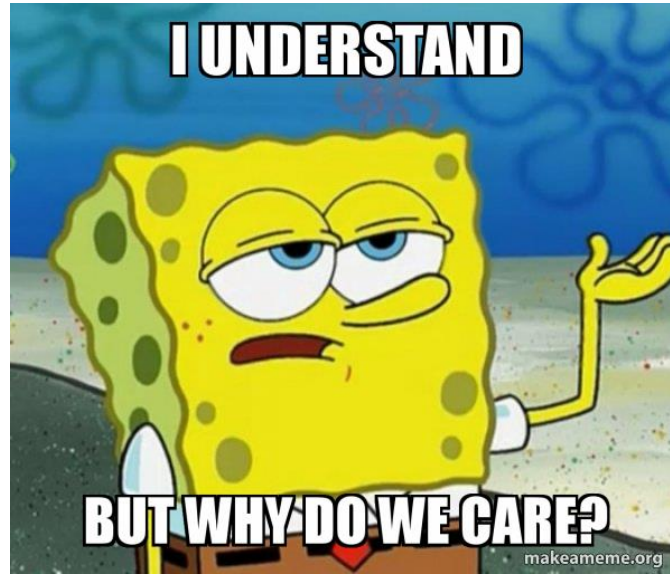
Introduction – Language model

$$P(w_0, w_1, \dots, w_m) = \prod_{i=0}^m P(w_i | w_0, w_1, \dots, w_{i-1})$$

- ❖ A Language Model (LM) is a probability distribution over sequences of words.
- ❖ Goal is to predict the probability of the next token (w_i) given the history of previous tokens ($w_{1:i-1}$).

Introduction – Language model

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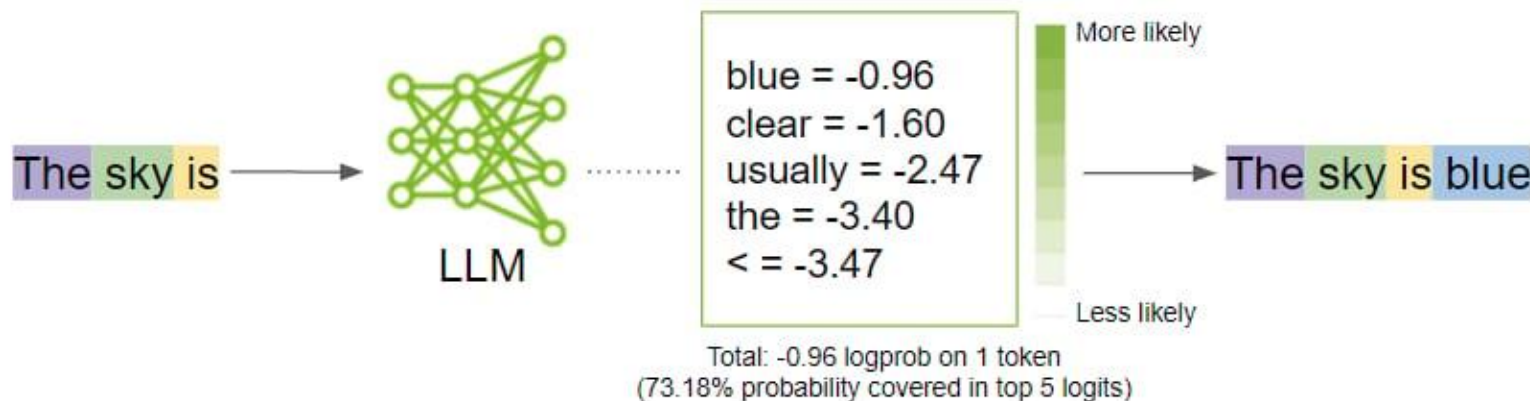
Answer Generation is a special case
of language modeling

$$P(w_0, w_1, \dots, w_m | c) = \prod_{i=0}^m P(w_i | w_0, w_1, \dots, w_{i-1}, c)$$

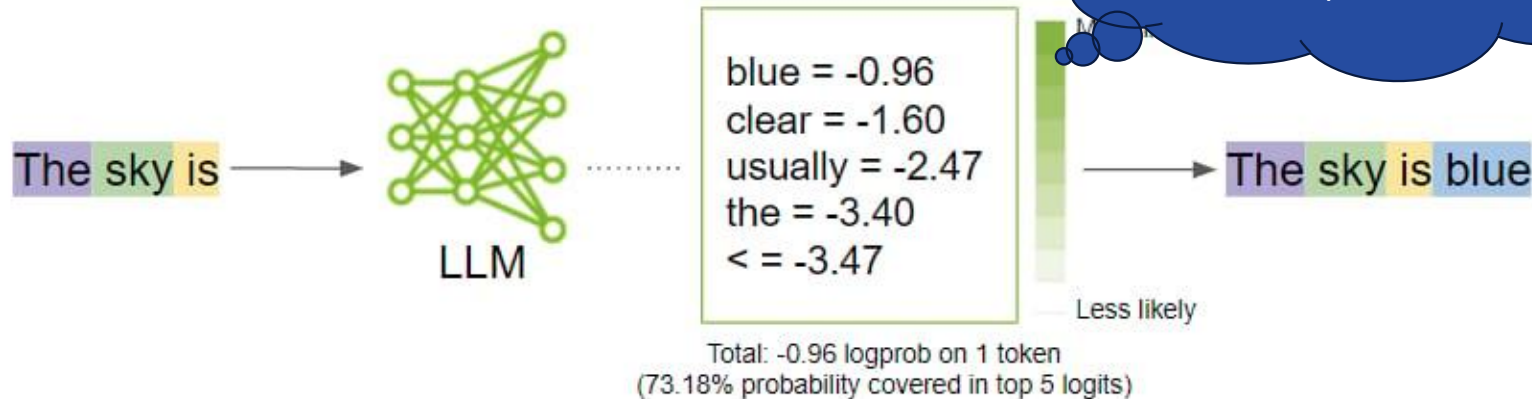
Introduction – Language model

$$P(w_0, w_1, \dots, w_m) = \prod_{i=0}^m P(w_i | w_0, w_1, \dots, w_{i-1})$$

- ❖ Goal we want to approximate the conditional probability of the next word given a context of words.
 - Neural networks, namely transformer-based architectures are the SOTA for this task.

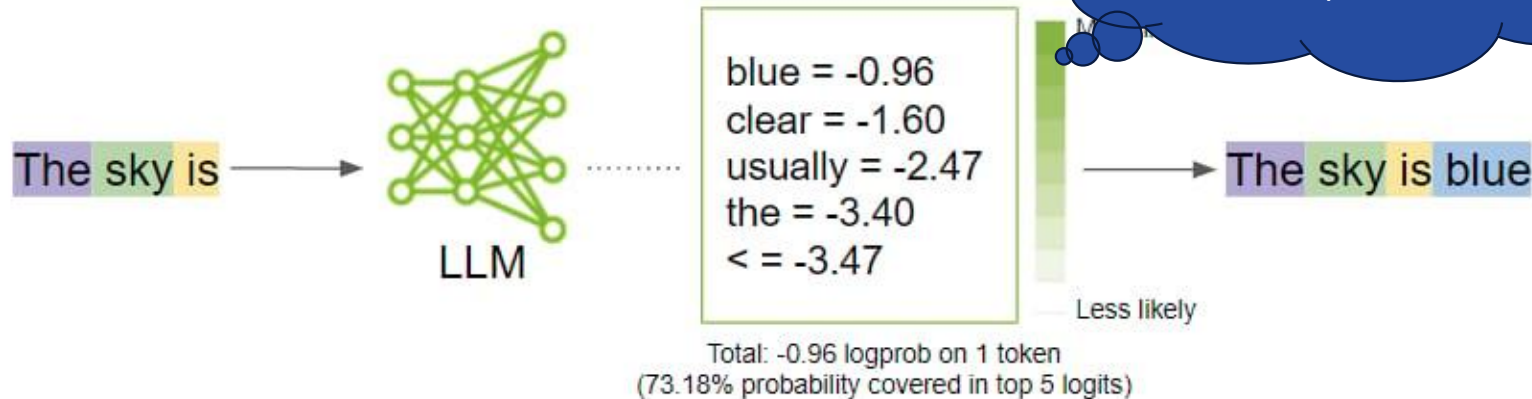


Introduction – Language model



Why do we use log-probs? Instead of probs?

Introduction – Language model



❖ Numeric stability!

- What happens when we have a chain of multiplications? The number will underflow!
- Log transform smaller numbers in big numbers and multiplication in additions!

Introduction – How to generate text?

- ❖ How can we generate text from a probabilistic model?

Introduction – How to generate text?

❖ How can we generate text from a probabilistic model?

- By sampling

$$\hat{y} = \arg \max_{\{w_0, w_1, \dots, w_m\}} P(w_0, w_1, \dots, w_m | c) = \arg \max_{\{w_0, w_1, \dots, w_m\}} \prod_{i=0}^m P(w_i | w_0, w_1, \dots, w_{i-1}, c)$$

❖ Greedy search

- Always selects the single most probable token.

❖ Beam Search

- Selects the most probable chain of tokens within a “beam” width.

❖ Random Sampling

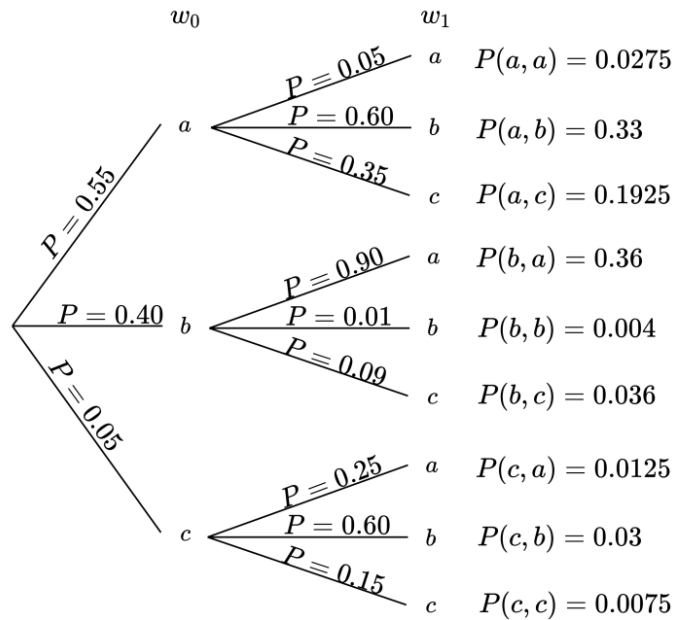
- Randomly samples the next token based on the probability distribution.

❖ Contrastive Search [1]

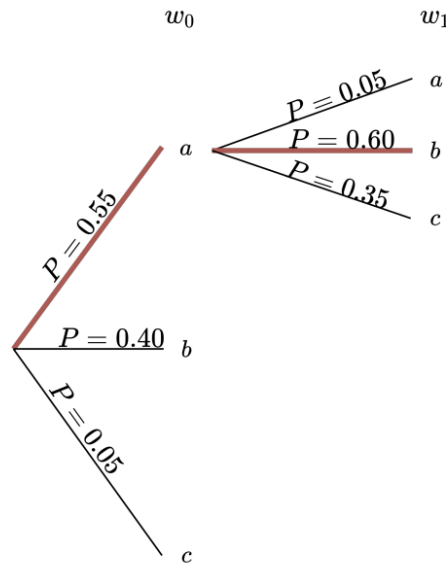
- A deterministic method that penalizes the model's top predictions if they are too similar to the history, enforcing diversity without sacrificing coherence.

Introduction - Sampling

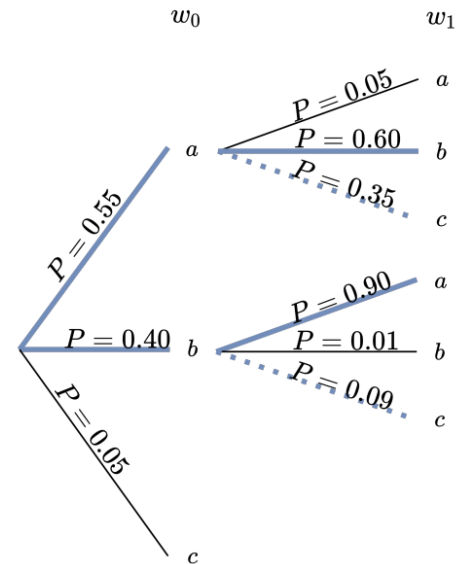
Brute Force



Greedy Search



Beam Search ($k = 2$)



Introduction - Sampling

$$x_t = \arg \max_{v \in V^{(k)}} \left\{ (1 - \alpha) \times \underbrace{p_\theta(v | \mathbf{x}_{<t})}_{\text{model confidence}} - \alpha \times \underbrace{(\max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\})}_{\text{degeneration penalty}} \right\}$$

prefix	Butt criticized Donald's controls in certain situations in the game, as well as the difficulty of some levels and puzzles. Buchanan also criticized the controls, calling them " too easy " and " too difficult ". He also criticized the game's graphics, stating that the game's graphics were " too simplistic " and " too simplistic ". He also criticized the game's graphics, stating that the game's graphics were " too simplistic " and " too simplistic ". He also criticized the graphics, stating that the game's graphics were " too simplistic " and " too simplistic ". He also criticized the graphics, stating that the game's graphics were " too simplistic " and " too simplistic ". He also criticized the graphics, stating that the game's graphics were " ...	div.	coh.	gen-ppl
beam	them " offensively rudimentary ". After the game was released, Buchanan wrote a number of game reviews about his review of the game, primarily summarizing what he thought the reviewers had enjoyed about it. Buchanan was pleased with how consistently these reviews awarded the game. For one, he praised the puzzles, compared them to Nos. 1 and 2 and enjoyed the construction of the artificial intelligence in the game's forms. He found no fault with the puzzles, even in those situations where they seemed repetitive. According to an Xbox Magazine review, " every time ...	0.02	0.605	7.91
nucleus	them " unimpressive " and a " nightmare " of an experience to play with players unfamiliar with Tetris. On the other hand, his opinion was shared by other reviewers, and some were critical of the game's technical design for the Wii version of Tetris. In addition, Tintin's review included a quote from Roger Ebert, who said that Tetris was better than the original game due to its simplicity and ease of play. Ebert's comments were included in the game's DVD commentary, released on March 22, 2010. It is unclear if any of the video commentary was taken from ...	0.98	0.597	37.83
contrastive		0.98	0.626	19.64

What are the main problems with current LLMs? (GAP)

❖ Hallucinations

- The probabilistic model will sample a plausible-sounding answer even if it's factually wrong.

❖ Knowledge Cutoff:

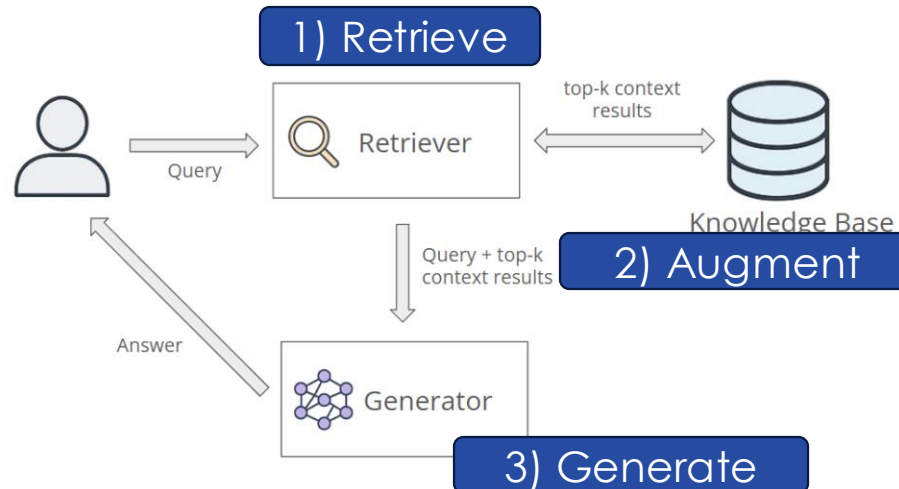
- The LLM becomes frozen after training. It knows nothing of events that happened post-training.

❖ Private Data:

- The model “cannot access” (hopefully) your data.

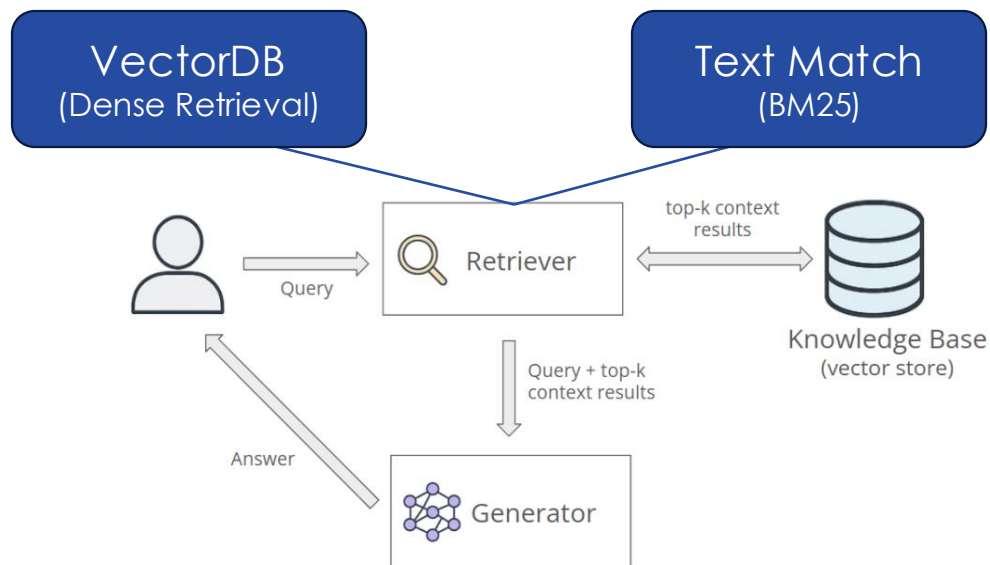
RAG – Retrieval Augmented Generation

- ❖ A framework that grants the LLM access to external data during the generation process.
- ❖ Moving from **Parametric Memory** (weights) to a **Non-Parametric Memory** (external index).



RAG != VectorDB

- ❖ RAG is a **framework**, not a specific technology.
 - You can perform retrieval using many different tools. The goal is simply to find relevant context.



How VectorDB works?

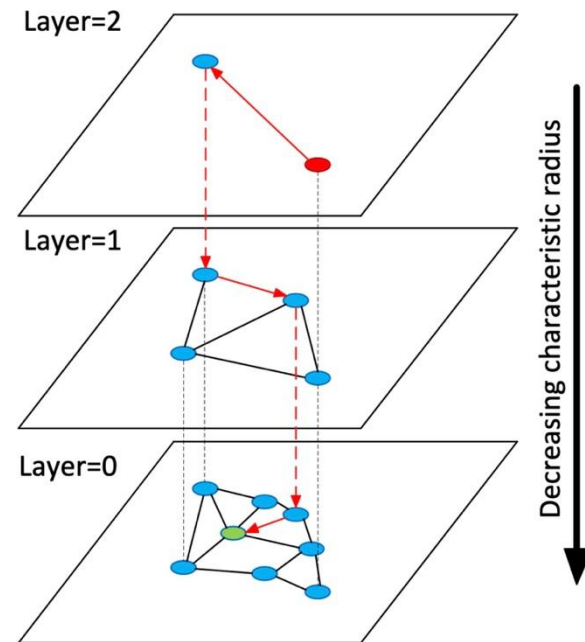
- ❖ You already know how text-based search engine works, lest explore a vectorDB now.

$$\vec{q} \cdot D$$

- ❖ Core of any VectorDB is the inner-product between a query and a document collection.
 - If the document collection is large this would be slower for real-time applications.
- ❖ How can we scale this?

HNSW

- ❖ Hierarchical Navigable Small World
 - Widely adopted in most commercial vectorDB
- ❖ Approximate Nearest Neighbor algorithm.
 - Extremely fast but it may fail to find the perfect answer.
 - It can skip 99% of the data.
 - Logarithmic complexity.
- ❖ It can skip 99% of the data



Literature examples

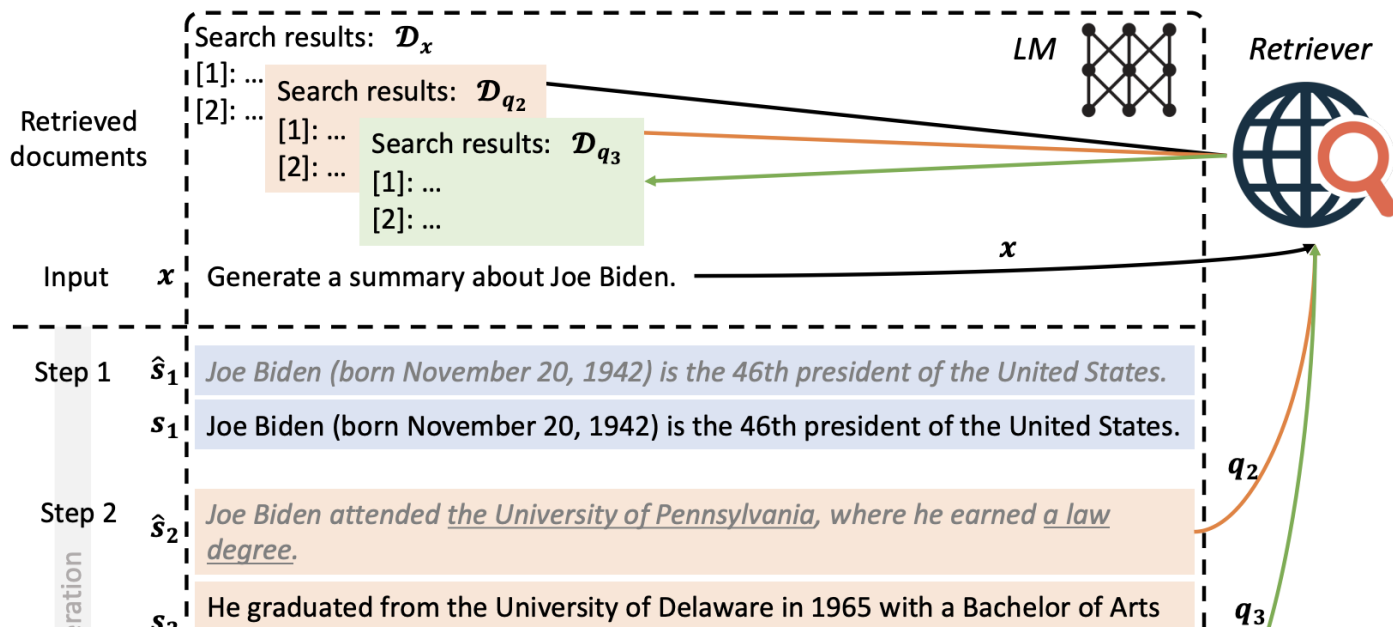
❖ FLARE

❖ ITER-RETGEN

❖ Self-RAG

FLARE

- ❖ The model decides when to retrieve. It starts generating; if it feels "uncertain" (low probability logits), it pauses, retrieves more info, and continues.



ITER-RETGEN

- ❖ The model uses the retrieved information to generate a partial answer, then uses that partial answer to search for *more* specific information in a loop.

Question: q

What is the height of the player who won the 2015 AFL Rising Star award?

Retrieval: $q \rightarrow \mathcal{D}_q$

Title: 2015 AFL Rising Star Context: The NAB AFL Rising Star award is given annually to a stand out young player in the Australian Football League. The award was won by Jesse Hogan of Melbourne

Retrieval-Augmented Generation: $\mathcal{D}_q \parallel q \rightarrow y_1$

The 2015 AFL Rising Star award was won by Jesse Hogan of Melbourne. Jesse Hogan is a professional Australian rules footballer. He is 198 cm tall. So the answer is 198 cm

Iteration 1

Question: q

What is the height of the player who won the 2015 AFL Rising Star award?

Generation-Augmented Retrieval: $y_1 \parallel q \rightarrow \mathcal{D}_{y_1 \parallel q}$

Title: Jesse Hogan Context: Jesse Hogan ... playing for the Melbourne Football Club. A key forward, Hogan is 1.95 m tall ... made his AFL debut in the 2015 season and won the Ron Evans Medal as the AFL Rising Star

Retrieval-Augmented Generation: $\mathcal{D}_{y_1 \parallel q} \parallel q \rightarrow y_2$

The 2015 AFL Rising Star award was won by Jesse Hogan of Melbourne. Jesse Hogan is 1.95 m tall. So the answer is 1.95 m

Iteration 2

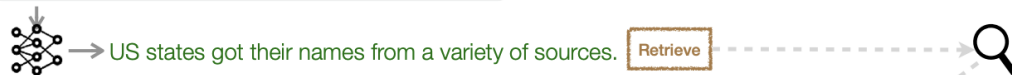
Self-RAG

- ❖ Self-RAG is a framework where the model **critiques itself** during the generation process using special "reflection tokens".

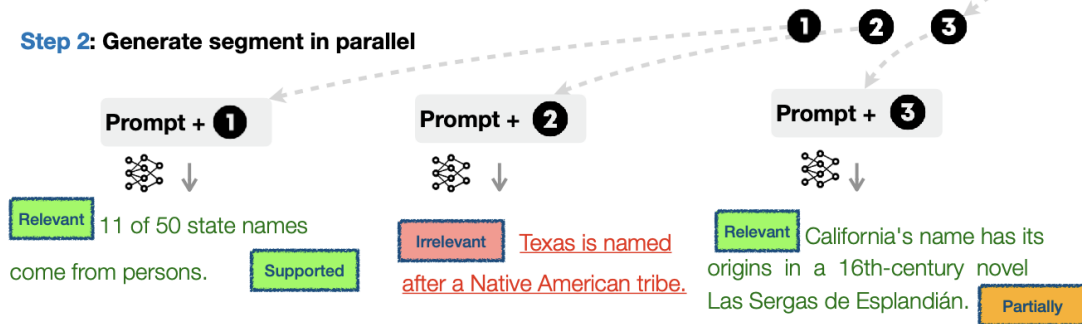
Ours: Self-reflective Retrieval-Augmented Generation (Self-RAG)

Prompt How did US states get their names?

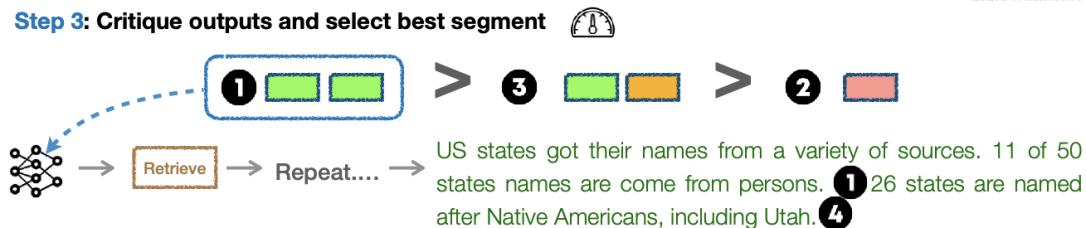
Step 1: Retrieve on demand



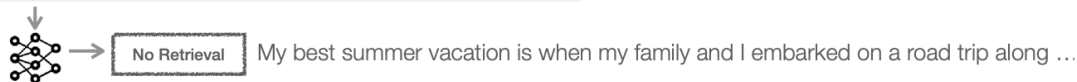
Step 2: Generate segment in parallel



Step 3: Critique outputs and select best segment



Prompt: Write an essay of your best summer vacation



Next steps

❖ Agentic Retrieval-Augmented Generation [1]

- Less rigid framework
- Reasoning-driven